

COVID-19 Severity Prediction

A machine learning project using chest CT features

Team: JASHRUTH K A - PES2UG24AM069

KISHAN BHARADWAJ - PES2UG24AM074

Problem statement

The project aims to classify COVID-positive patients as having severe or non-severe outcomes using information extracted from chest CT scans. Rather than predicting ICU admission, mechanical ventilation, and death separately, this implementation uses one binary outcome: intubation or death within one month. The objective is to build and evaluate an interpretable educational ML pipeline, not a clinical decision system.

Dataset details

The project uses the publicly accessible STOIC2021 training dataset. It contains 2,000 three-dimensional CT scans with COVID and severity labels, including 1,205 COVID-positive patients. Only COVID-positive patients were eligible for this project. Scans are downloaded as Metalmage (.mha) volumes, and labels are read from reference.csv. The dataset is distributed under CC BY-NC 4.0 terms.

Two patients were used only for initial inspection and excluded from modeling. A reproducibly sampled cohort of 100 patients contained 75 non-severe and 25 severe outcomes. A stratified split allocated 80 patients to training and 20 to initial testing. After tuning, a separate group of 40 previously unused patients (30 non-severe and 10 severe) provided the final internal evaluation.

Approach

A pretrained lungmask R231 model segments the lungs in each CT volume. Left and right lung labels are combined into one mask so that features are measured within the lungs. Selected CT slices and mask overlays are inspected visually. The segmentation model is a preprocessing tool; logistic regression is the single classifier that predicts the patient outcome.

Eleven features summarize each patient: lung volume; mean, standard deviation, 10th percentile, median, and 90th percentile of intensity; histogram entropy; and texture contrast, homogeneity, energy, and entropy. Intensities are clipped to -1000 to 400 Hounsfield units. The custom 2D texture calculation uses 32 intensity bins and 1 mm in-plane resampling. This is a compact radiomics baseline rather than a full standardized 3D radiomics implementation.

Features are standardized inside a scikit-learn pipeline and supplied to logistic regression. Five-fold stratified cross-validation on training patients compares regularization strengths and class weighting, selecting settings by balanced accuracy. The selected model uses $C = 0.1$ and balanced class weights, with a fixed decision threshold of 0.5.

Implementation overview

The workflow is implemented in a Python notebook on Google Colab using a Tesla T4 GPU for lung segmentation. SimpleITK reads CT volumes, NumPy and SciPy compute features, pandas manages patient tables, scikit-learn trains and evaluates the classifier, and Matplotlib generates the result plots. Scaling is fitted within each training fold to prevent preprocessing leakage. Patient IDs and outcome labels are excluded from the predictors.

Scans, masks, extracted features, patient selections, and model files are saved to Google Drive. Feature tables are updated after every completed patient, allowing interrupted processing to resume. The tuned classifier and its feature order are saved using joblib. For evaluation, the saved classifier receives features from the 40 new patients without retraining. Predictions are compared against their known outcomes; metrics and plots are saved for reporting. Processing the final 40 scans took 27.9 minutes, while classifier training and inference were much faster.

Results

On the 40 previously unused patients, the tuned model correctly classified 31 patients. The table summarizes the final evaluation at the fixed threshold:

Metric	Result
Accuracy	77.5%
Balanced accuracy	71.7%
ROC-AUC	0.743
Average precision	0.619
Severe-case recall	60.0%

The confusion matrix contains 25 correctly identified non-severe patients, 5 false severe predictions, 6 correctly detected severe patients, and 4 missed severe patients. Predicting every patient as non-severe would achieve 75% accuracy but detect no severe cases. Balanced accuracy and severe recall therefore provide essential context for the headline accuracy.

Conclusions and limitations

The project demonstrates an end-to-end pipeline from CT preprocessing to outcome prediction and evaluation. The model shows moderate discriminatory ability and detects 6 of 10 severe cases in the final sample. However, it still misses severe outcomes, and only 10 severe evaluation patients make the estimate uncertain. The original and tuned models were tested on different groups, so their score difference cannot establish the effect of tuning alone.

The final evaluation comes from the same dataset and is not external validation. Masks were inspected on selected slices without expert or quantitative validation. Texture extraction does not standardize slice spacing, and prediction scores are not calibrated clinical risks. More training data, stronger segmentation checks, and independent evaluation could improve the evidence. The current implementation is suitable as an educational prototype, with these limitations explicitly reported.

Sources: STOIC2021: stoic2021.grand-challenge.org/stoic-db/; lungmask: github.com/JoHof/lungmask; Hofmanninger et al. (2020): doi.org/10.1186/s41747-020-00173-2. Results are from the executed project notebook.